

HTML Forms (Beginner Friendly)

What is a Form?

A **Form** is an HTML element that allows users to enter information and send it to a web server.

Examples

- Login Form
- Registration Form
- Contact Form
- Job Application Form
- Feedback Form
- Online Exam Form

Whenever a website asks you to type something, you are using a **form**.

Example

```
Name      : _____  
Email     : _____  
Password  : _____  
[ Submit ]
```

This entire section is called a **Form**.

Why Do We Need Forms?

Forms are used to collect information from users.

Examples

- Student Registration
- Employee Registration
- COVID-19 Registration

Book a ticket for a train

- Bank Account Opening
- Online Shopping Checkout
- Hotel Booking
- Railway Reservation

Without forms, websites cannot receive information from users.

The `<form>` Tag

Every HTML form starts with the `<form>` tag.

Syntax

```
<form>

  <!-- Form controls go here -->

</form>
```

Everything inside the `<form>` tag belongs to that form.

Example

```
<form>

  Name:
  <input type="text">

</form>
```

Form Controls

The elements placed inside a form are called **Form Controls**.

Some common form controls are:

- Text Box
- Password Box
- Radio Button
- Checkbox

- Dropdown List
- Date Picker
- Text Area
- Buttons
- File Upload
- Range Slider

The `<label>` Tag

The `<label>` tag tells the user what information should be entered.

Example

```
<label>Name</label>
<input type="text">
```

Output

Name _____

The `<input>` Tag

The `<input>` tag is the most commonly used HTML form element.

Syntax

```
<input>
```

Different values of the `type` attribute make the `<input>` element perform different tasks.

Common Input Types

1. Text

1. Text

Used to enter normal text.

Syntax

```
<input type="text">
```

Uses

- Name
- City
- College
- Country

2. Password

Used to hide typed characters.

Syntax

```
<input type="password">
```

Output

```
*****
```

3. Email

Used to enter email addresses.

Syntax

```
<input type="email">
```

Example

4. Number

Accepts only numeric values.

Syntax

```
<input type="number">
```

Uses

- Age
- Salary
- Marks
- Mobile Number

5. Date

Displays a calendar for selecting dates.

Syntax

```
<input type="date">
```

Uses

- Date of Birth
- Joining Date
- Vaccination Date

6. Radio Button

Allows the user to select **only one** option.

Syntax

```
<input type="radio">
```

Example

Gender

() Male
() Female

7. Checkbox

Allows the user to select **multiple** options.

Syntax

```
<input type="checkbox">
```

Example

Hobbies

☐ Music
☐ Sports
☐ Travel
☐ Movies

8. File Upload

Allows the user to upload a file.

Syntax

```
<input type="file">
```

Uses

- Resume
- Photo
- Certificates

- Certificates

9. Range Slider

Creates a slider.

Syntax

```
<input type="range">
```

Uses

- Income
- Volume
- Brightness

The `<textarea>` Tag

The `<textarea>` element allows users to enter multiple lines of text.

Syntax

```
<textarea></textarea>
```

Unlike the `<input>` element, the `<textarea>` element has both an opening and a closing tag.

Uses

- Address
- Bio
- Feedback
- Comments

Dropdown List (`<select>`)

Dropdown lists are created using the `<select>` element.

Syntax

```
<select>
  <option>India</option>
  <option>USA</option>
  <option>Canada</option>
</select>
```

Uses

- Country
- Degree
- Passing Year
- Vaccination Center

Buttons

Buttons perform actions.

Common button types are:

- Submit
- Reset
- Normal Button

Examples

```
<input type="submit">
<input type="reset">
<button>Click Here</button>
```

The **placeholder** Attribute

The **placeholder** attribute displays a hint inside an input field.

Example

```
<input type="text" placeholder="Enter your name">
```

Output

```
[ Enter your name ]
```

The name Attribute

The `name` attribute gives a name to the form field.

Example

```
<input type="text" name="fullname">
```

When the form is submitted, the data is sent as:

```
fullname = Rahul
```

The value Attribute

The `value` attribute specifies the default value.

Example

```
<input type="text" value="India">
```

Output

```
India
```

The checked Attribute

The `checked` attribute is used with radio buttons and checkboxes.

Example

```
<input type="radio" checked>
```

The radio button is selected by default.

The **selected** Attribute

The **selected** attribute is used inside the **<option>** element.

Example

```
<select>
  <option>USA</option>
  <option selected>India</option>
  <option>Canada</option>
</select>
```

The option **India** appears selected by default.

The **required** Attribute

The **required** attribute makes a field compulsory.

Example

```
<input type="text" required>
```

The form cannot be submitted until the field is filled.

The **readonly** Attribute

The **readonly** attribute allows the user to view the value but not modify it.

Example

```
<input type="text" value="India" readonly>
```

The disabled Attribute

The `disabled` attribute completely disables a form control.

Example

```
<input type="text" disabled>
```

The user cannot interact with this field.

The <fieldset> Tag

The `<fieldset>` tag is used to **group related form elements** together inside a form.

It draws a box around the grouped elements, making the form more organized and easier to understand.

It is especially useful for large forms where fields can be divided into sections such as:

- Personal Information
- Educational Information
- Address Details
- Job Details
- Payment Information

Syntax

```
<fieldset>  
  Form Elements  
</fieldset>
```

Example

```
<form>

  <fieldset>

    <label>Name</label>
    <input type="text">

    <br><br>

    <label>Email</label>
    <input type="email">

  </fieldset>

</form>
```

Output

```
+-----+
|
| Name   _____
|
| Email  _____
|
+-----+
```

The border around the fields is created automatically by the browser.

The `<legend>` Tag

The `<legend>` tag is used to **provide a title or caption** for a `<fieldset>` .

It helps users understand what type of information is contained inside that group.

The `<legend>` tag must always be placed **inside** the `<fieldset>` tag and is usually the first element.

Syntax

```
<fieldset>
```

```
<legend>Personal Information</legend>
```

```
Form Elements
```

```
</fieldset>
```

Example

```
<form>

  <fieldset>

    <legend>Student Information</legend>

    <label>Full Name</label>
    <input type="text">

    <br><br>

    <label>Email</label>
    <input type="email">

  </fieldset>

</form>
```

Output

```
+-----+
| Student Information |
+-----+
| Full Name  |_____|
+-----+
| Email      |_____|
+-----+
```

The text **Student Information** appears as the title of the fieldset.

Difference Between `<fieldset>` and `<legend>`

<code><fieldset></code>	<code><legend></code>
Groups related form controls together.	Provides a title or caption for the fieldset.
Creates a border around the grouped elements.	Displays the heading on the border of the fieldset.
Used for organizing large forms.	Helps identify the purpose of the grouped fields.